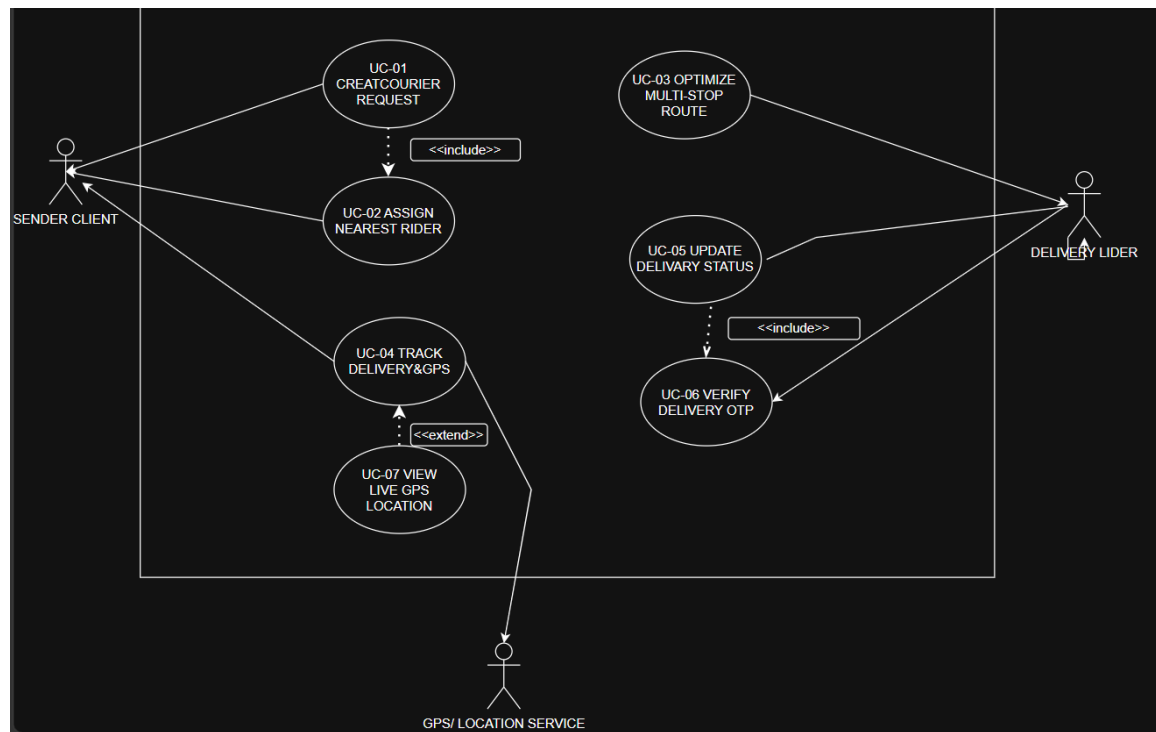


Requirements Table – Hyperlocal Courier Dispatch & Tracking Engine

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match an outgoing courier request with the nearest active delivery rider within a 3 km radius.	High	Pass: An active rider within 3 km receives the dispatch notification. Fail: Order is assigned to an offline rider or rider beyond 3 km.	Ensures fast local pickup.
FR-002	Functional	The system shall allow the sender client to create a courier request by entering pickup location, destination, and parcel details.	High	Pass: A valid courier request is created and assigned a unique order ID. Fail: Request is created with missing mandatory information.	Enables parcel booking.
FR-003	Functional	The system shall provide the delivery rider with an optimized route for completing multiple parcel stops.	High	Pass: The rider receives an ordered route containing all assigned stops. Fail: Any assigned stop is missing from the route.	Reduces delivery time and travel distance.
FR-004	Functional	The system shall allow the sender client to track the rider's current delivery status and GPS location.	High	Pass: Updated rider location and delivery status are displayed to the sender. Fail: Location/status is not updated.	Provides real-time delivery visibility.
FR-005	Functional	The system shall require the delivery rider to enter a valid OTP before marking a parcel as delivered.	High	Pass: Correct OTP marks the parcel as delivered. Fail: Incorrect OTP prevents delivery completion.	Prevents unauthorized delivery confirmation.
NFR-001	Nonfunctional	The system shall transmit delivery status and live GPS telemetry updates to the sender client with under 2-second latency.	High	Pass: 95% or more of updates reach the sender within 2 seconds during load testing. Fail: Target latency is exceeded.	Provides responsive real-time tracking.
NFR-002	Nonfunctional	The system shall protect courier and location data using authenticated and encrypted communication.	High	Pass: Unauthorized users cannot access delivery/location data and communication uses encryption. Fail: Protected data is accessible without authentication.	Protects sensitive delivery information.



Actor: Sender Client

Supporting Actor: Delivery Rider

Preconditions

1. Sender Client is registered/logged into the system.
2. Location services are available.
3. Delivery riders are available in the service area.

Postconditions

1. Courier request is successfully created.
2. A nearby active rider is assigned.
3. Rider receives pickup details.
4. Sender receives assignment confirmation.

Main Success Scenario

1. Sender opens the courier delivery application.
2. Sender enters the pickup location.
3. Sender enters the destination location.
4. Sender enters parcel details.
5. System validates the entered information.
6. System creates a courier request and generates an order ID.

7. System searches for active delivery riders near the pickup location.
8. System identifies the nearest active rider within a 3 km radius.
9. System sends the pickup and delivery details to the rider.
10. Rider accepts the delivery request.
11. System confirms the rider assignment to the sender.
12. Use case ends successfully.

Alternate Flow

A1. No rider available within 3 km

1. System searches for available riders within the defined radius.
2. No suitable rider is found.
3. System informs the sender that no rider is currently available.
4. Sender can retry the request later.
5. The courier request remains pending.